Critical evaluation of the modified Gompertz response function as a parametric template for treatment-response trajectories in N-of-1 trial simulation

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Contents

1 Abstract

Background. Simulation-based power analyses for aggregated N-of-1 trials require a parametric template for the within- participant response trajectory. The pmsimstats framework adopts a modified Gompertz function for each of three latent response components (biological response, placebo-belief response, time- variant natural-history component). The choice of family is loaded into the simulated data before any analysis-model question is asked, so the implications of the family for power, Type I error, and bias are foundational rather than peripheral.

Methods. We evaluate the modified Gompertz family against the relevant biological, pharmacokinetic, and biostatistical literature. We characterize the parameter space the function covers, the trajectories it can and cannot represent, and the sensitivity of standard pmsimstats inference to misspecification of the response-curve family. Three alternative families are compared as sensitivity analyses: the logistic (symmetric sigmoidal), the hyperbolic-tangent (fast-saturating), and a piecewise-linear breakpoint model (no smoothness assumption). For each alternative we compute power, Type I error, and bias of the biomarker-treatment interaction estimate at trial-relevant sample sizes against simulated data drawn from the alternative family but analyzed under the standard linear-mixed model.

Results. A 1{,}000-replicate-per-cell simulation across 16 cells (4 trajectory families $\times$ 2 data-generating process (DGP) architectures $\times$ 2 biomarker effect-size levels) at $N = 70$, open-label plus blinded discontinuation (OL+BDC) design, $t_{1/2} = 0$, found that the inferential consequence of trajectory family is architecture-dependent. Under the covariance-moderation architecture (multivariate normal (MVN) differential correlation), where the biomarker-treatment interaction lives in the covariance structure and is therefore mean-independent, trajectory family is exactly inert: with matched random-number seeds the four families return identical rejection decisions, so the production-seed spread $\{0.774, 0.777, 0.791, 0.804\}$ is sampling noise. Under the mean-moderation architecture, implemented in its faithful multiplicative form in which the biomarker moderation rides the response trajectory, the four families separate: powers were $\{0.743, 0.760, 0.778, 0.782\}$ for Gompertz, hyperbolic-tangent, piecewise-linear, and logistic respectively, a 0.039 spread in which the Gompertz default sits roughly two Monte Carlo standard errors (MCSE) below the logistic and is the lowest-power, that is mildly conservative, family. Type I error at the null sat in $[0.019, 0.038]$ across all eight null cells, slightly conservative and without family-specific inflation; bias of the interaction estimate was negligible and family-invariant (≈ -0.23).

Conclusions. The modified Gompertz family is a defensible default for the response-curve template in pmsimstats N-of-1 simulation, with two qualifications. Under a covariance-channel (MVN) data-generating process, trajectory family cannot affect the interaction test by con-

struction, so the choice is inferentially free. Under a mean-moderation data-generating process, trajectory family does carry a modest inferential cost at the few-percent level, and the Gompertz default is mildly conservative relative to the symmetric and breakpoint alternatives. We recommend reporting Gompertz-vs-alternative sensitivity at least once per parameter regime where simulation results inform protocol-design decisions, with particular attention to mean-moderation regimes, and we propose a default sensitivity grid that the framework can run alongside its primary cells.

2 Introduction

2.1 A claim about response-curve choice in N-of-1 trial simulation

The first claim is that the modified Gompertz function used as the parametric template for treatment-response trajectories in aggregated N-of-1 trial simulation (specifically, in the pmsimstats simulation framework that operationalizes the Hendrickson et al. design family[1]) is a defensible default for the saturating-shape regime that characterizes most pharmacodynamic and chronic-condition response curves over the typical trial timescale. The function is well-established in tumor growth dynamics[2,3], microbial growth modeling[4], and the broader unified-Richards growth-curve family[5–7]; its three-parameter form is flexible enough to capture the asymmetric saturation profiles that linear extrapolation, exponential decay, or symmetric logistic models cannot.

The second claim is that family-misspecification sensitivity is essentially absent from the published N-of-1 methodology literature, and that it should be reported routinely rather than assumed away. The methodological N-of-1 corpus[1,8–10] is, to the best of our knowledge, silent on parametric trajectory shape: papers either inherit a chosen family from the upstream literature without comparing alternatives, or specify a linear or AR(1) form that is too coarse to capture the saturation that real trajectories exhibit. The oncology-modeling literature has set the precedent for what a proper family-comparison sensitivity analysis looks like[6,11], and the pharmacology literature has converged on Hill / sigmoid-Emax / indirect-response models for related questions[12,13]. The N-of-1 literature has not imported this comparative apparatus, and the present paper endeavors to supply it.

The two claims are coupled because the case for a chosen family is incomplete without the comparison. We do not argue that Gompertz is the correct family in any strict sense; rather, we argue that it is a reasonable choice within the unified-Richards family of saturating curves, and that the inferential consequences of choosing it (versus the symmetric logistic, the von Bertalanffy growth curve[14], or a piecewise-linear breakpoint model) should be quantified rather than asserted. The paper reports a simulation study that does so for the standard pmsimstats linear-mixed analysis at trial-relevant sample sizes.

2.2 Why the question matters

First, the chosen family loads into the data before any analysis-side question is asked. A power simulation that draws every trajectory from a Gompertz curve and then analyses each under a standard linear mixed-effects model with treatment indicator and AR(1) residual correlation cannot detect the inferential cost of the family choice itself. Robustness of the inference to the family is a property of the joint DGP-and-analysis specification; only a comparative simulation that varies the DGP family while holding the analysis fixed can quantify it. This is the same logic that underpins analysis-model misspecification studies in the broader clinical-trial methodology literature[15], applied one level down in the simulation hierarchy, and it appears to us that the N-of-1 simulation literature has not yet drawn the parallel explicitly.

Second, the function family has known identifiability fragility in short noisy trajectories. Joint estimation of the three Gompertz parameters (m, d, r) in $y = m\exp(-d\exp(-rt))$ is poorly conditioned: the asymptote and rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region.[6] and[7] propose unified-Richards reparameterizations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote) with lower correlation, and[16] develops the formal 'sloppiness' diagnostic that quantifies this directly as a property of the parameter space.[11] fitted eight tumor-growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalized logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories, and its absence is, in our view, problematic.

Third, the default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via the broader growth-curve tradition, where its use traces to Laird's tumor dynamics work[2] and to Zwietering's microbial growth reformulations[4]. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family[12,17] and to Jusko's indirect-response models[13]. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defense against the pharmacology default. Either default may be adequate for the saturation regime, but the inheritance is not itself evidence, and the question merits a direct empirical answer.

2.3 What the published N-of-1 literature does and does not do

A review of the recent N-of-1 methodology literature confirms, to the best of our knowledge, a silence on the parametric-trajectory-family question.

The CONSORT extension for N-of-1 trials[10] codifies reporting standards but does not address parametric trajectory shape. The foundational programmatic statements[8,9] emphasize design and analysis broadly without committing to a parametric response form. The Hendrickson et al. framework[1] introduces the modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers in chronic-condition N-of-1 trials (including the prazosin-PTSD reference work building on[18]) inherit the framework without examining the choice. Unfortunately, this pattern of silent inheritance is pervasive enough that no published N-of-1 simulation study, to the best of our knowledge, has broached the family-misspecification question.

In adjacent literatures the comparative apparatus does exist. The oncology-modeling community routinely fits multiple growth families and compares them on identifiability and predictive performance[11]. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models[19,20]. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modeling[21]. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

The present paper's contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis, with results reported in the Morris-White-Crowther[15] ADEMP framework already adopted as the project-wide simulation-reporting standard.

2.4 What this paper contributes

In section 2 we characterize the modified Gompertz family as implemented in pmsimstats: the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterization[6,7] that supplies more interpretable and less correlated parameters.

In section 3 we survey three alternative response-curve families: the symmetric logistic[22], the hyperbolic-tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

In sections 4 and 5 we conduct a Monte Carlo sensitivity-analysis simulation that generates trajectories under each of the four families and analyses them under the standard pmsimstats linear-mixed model. We report power, Type I error, and bias of the biomarker-treatment interaction estimate under each cell, with conventional MCSEs. The simulation borrows the comparative methodology of[11] (eight-model tumor-growth comparison). The sloppiness diagnostic of[16] motivates the identifiability framing in this Introduction but is not itself computed in the present 16-cell focused submission; it is Study 2 of the ADEMP pre-registration (`00-ademp-pre-registration.md`) and remains future work, not a result reported in Sections 4-5 below.

In section 6 we identify the trajectory regimes in which Gompertz misspecification would be expected to be consequential: non-saturating growth beyond the trial window, biphasic patterns, and breakpoint behavior. For these regimes, the standard N-of-1 inference is materially affected by the family choice, and we recommend specific sensitivity-grid extensions that pmsimstats users should run alongside their primary cells.

The full mathematical and historical context, including citations that did not survive editing into the manuscript proper, is preserved in the companion technical document at `docs/25-gompertz-model-evaluation.md`.

3 The modified Gompertz function in pmsimstats

The Gompertz curve originates in Gompertz's 1825 law of human mortality[23] and was first applied as a general growth curve by[24]; the pmsimstats framework adopts the modified, monotone-saturating form of that curve. The framework parameterizes each of the three latent response components (biological response BR, placebo-belief response PB, and time-variant natural-history component TV) with a modified Gompertz function

$$y(t) = \mathrm{maxr} \cdot \exp(-\mathrm{disp} \cdot \exp(-\mathrm{rate} \cdot t)), \tag{1}$$

with vertical-offset adjustment so that $y(0) = 0$. The three parameters are maxr (asymptotic maximum response, matching the on-drug or on-natural-history saturation level), rate (which controls how quickly the response approaches its asymptote), and disp (which controls the curvature near $t = 0$). The Gompertz form is monotone-saturating and asymmetric: the inflection occurs at $t^* = \log(\mathrm{disp})/\mathrm{rate}$, with the rise being relatively slower before the inflection and faster after. The parameterization used at the prazosin/PTSD reference cell sets $\mathrm{maxr}_{BR} \approx 4$ nightmares per week, $\mathrm{rate}_{BR} \approx 0.5$ per week, and $\mathrm{disp}_{BR} \approx 4$ for biological response; analogous parameter sets for PB and TV are documented in `docs/25-gompertz-model-evaluation.md` and `R/utilities.R`.

4 Alternative response-curve families

Three alternative families serve as sensitivity-analysis comparators, and in this section we shall describe each in turn before the simulation study proper. Each is parameterized so that the asymptotic maximum and a single-anchor calibration point (the response level at $t = 5$ weeks, calibrated to the Gompertz family at the same anchor) are matched across families. That is, the families differ in shape envelope (smoothness, symmetry around the inflection, and curvature) while sharing the same marginal effect size at the anchor, a design choice that we discuss further in section 4.

4.1 Symmetric logistic

The logistic function

$$y(t) = \frac{\mathrm{maxr}}{1 + \exp(-\mathrm{rate} \cdot (t - t_0))} - \mathrm{offset} \tag{2}$$

is monotone-saturating and symmetric around its inflection at $t = t_0$. A vertical offset is subtracted so that $y(0) = 0$. At matched maxr and matched response level at the calibration anchor, the logistic family rises faster than Gompertz before the inflection and slower after, producing a visibly different shape envelope while sharing the same asymptote. The symmetric inflection is the principal qualitative contrast with the asymmetric Gompertz curve, and the reader will note that this symmetry property is, in a sense, the logistic family's most theoretically interesting feature for the purposes of the present comparison.

4.2 Hyperbolic-tangent

The hyperbolic-tangent variant

$$y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \tag{3}$$

is also monotone-saturating and symmetric, but with a qualitatively faster approach to the asymptote than the logistic. The tanh family produces the steepest pre-asymptote rise of the three smooth families, and it has been used in pharmacokinetic modeling for drugs with rapid receptor saturation. We include it here as the representative of the fast-saturating smooth regime that neither the logistic nor the Gompertz family can span with their standard parameterizations.

4.3 Piecewise-linear breakpoint

The piecewise-linear-breakpoint family, formalized in the broken-line regression literature[25],

$$y(t) = \text{maxr} \cdot \min(1,\, t/t_{\text{bp}}) + \text{post\_slope} \cdot \max(0,\, t - t_{\text{bp}}) \tag{4}$$

is the canonical non-smooth alternative: a linear ramp up to a breakpoint t_{bp}, then constant (when post_slope $= 0$) or with a small post-breakpoint slope. The breakpoint is the only non-differentiable feature across the four families and is the principal qualitative contrast with the three smooth families. This family is the strongest available test of the framework's sensitivity to non-smoothness in the trajectory. But how different is the piecewise-linear breakpoint from the smooth families, from the perspective of the Wald test that the analysis model applies? That question is precisely what the simulation in section 4 is designed to answer.

5 Sensitivity-analysis simulation design

The primary estimand is the biomarker-treatment interaction coefficient $\hat{\beta}_{bm:D}$ from the linear mixed-effects analysis model `Sx ~ bm + t + Dbc + bm:Dbc + (1|ptID)` with `corCAR1(form = ~t|ptID)`. Primary inferential outputs are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and Type I error under $\beta_{bm:D} = 0$. Secondary estimands are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true effect and the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates. Performance measures are reported with binomial-proportion MCSEs for proportions and the standard MCSE of the within-replicate mean for continuous quantities.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`, which fixes the four DGP families (modified Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study (Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

For the present submission the simulation is scoped to a 16-cell focused factorial: four trajectory families crossed with two DGP architectures (mean moderation and MVN differential correlation, following the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`) crossed with two biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-calibrated Hybrid OL+BDC, $N = 70$, $t_{1/2} = 0$, with carryover set to absent in order to isolate the trajectory-shape effect from carryover-mediated correlation decay. Following the compendium convention, N is the total across randomization paths: the driver allocates 35 participants to each of the design's two paths, and the paths are concatenated before analysis. Each cell is run at $n_{\text{reps}} = 1{,}000$ replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is deferred to a follow-up; the focused 16-cell factorial answers the central scientific question (does trajectory family choice matter for the `bm:Dbc` Wald test?) at MCSE 0.013–0.016 per cell.

Notation for the moderation parameter. Because this factorial crosses both architectures, c_{bm} is used here as a common label for the biomarker-moderation strength at a matched numeric value. Strictly, the parameter is the correlation c_{bm} under the covariance-moderation architecture and the dimensionless multiplier β_{bm} under the mean-moderation architecture, the two being calibrated to a common scale because the Architecture-A shift is applied as $\beta_{bm}\, b_i\, \sigma_{BR}$ on the standardized biomarker b_i. The companion architecture manuscript keeps the symbols distinct; we adopt the shared label only where a statement applies to both architectures at the matched value, and the architecture is named explicitly wherever a result is architecture-specific.

Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family's response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via uniroot (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver `analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R`.

Trajectory-family encoding. The family is routed through the data-generating process rather than imposed post hoc: it sets the BR mean curve inside the covariance construction via the `br_family` argument to `buildSigma`. Under the mean-moderation architecture the biomarker-treatment interaction is implemented in the multiplicative form of the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`, $Y_{it} = \mu_0 + \beta_1 D_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, so the biomarker-dependent component of the treatment effect rides the response trajectory $BR(t)$ rather than entering as a flat shift. We normalize the per-timepoint moderation so that its mean on-drug magnitude equals σ_{BR}, holding the marginal interaction effect size fixed across families while letting the temporal profile vary by family shape. Under the covariance-moderation architecture (MVN differential correlation) the interaction is encoded in the BM-BR covariance block, which is independent of the BR mean; the family therefore enters only through the mean curve and, as we verify by a matched-seed check, cannot move the interaction test.

6 Results

The simulation program was executed at 1{,}000 replicates per cell across the 16-cell trajectory-family-by-architecture-by- effect-size factorial in 597 s using 8-core `parallel::mclapply`. Convergence was 100% across all 16{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/07-gompertz-faithful-summary.txt`. Cell MCSE on power runs from 0.013 to 0.014 across the alternative cells and from 0.004 to 0.006 on the conservative null cells. We begin the results section with power, then turn to Type I error, and finally consider bias of the interaction estimate.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$, percent $\pm$ MCSE. The mean-moderation architecture is the faithful trajectory-scaled mean-moderation form; the covariance-moderation architecture is the MVN covariance channel:

Family	Mean	Covariance
Gompertz	74.3 ± 1.4	77.4 ± 1.3
Logistic	78.2 ± 1.3	79.1 ± 1.3
Hyperbolic tangent	76.0 ± 1.4	77.7 ± 1.3
Piecewise-linear breakpoint	77.8 ± 1.3	80.4 ± 1.3

The two architectures behave qualitatively differently. Under the covariance-moderation architecture (MVN differential correlation), the biomarker-treatment interaction is carried entirely by the covariance structure, which is independent of the BR mean curve; consequently the trajectory family, which sets that mean curve, cannot alter the interaction test. We confirm this directly: in a matched-seed check the four families return bit-for-bit identical rejection decisions, so the production-seed cell spread (0.774 to 0.804) is sampling noise rather than a family effect. Under the mean-moderation architecture, implemented in the faithful multiplicative form in which the biomarker-dependent component of the treatment effect rides the response trajectory, the families separate: power ranges from 0.743 (Gompertz) through 0.760 (hyperbolic-tangent) and 0.778 (piecewise-linear) to 0.782 (logistic), a 0.039 spread. The Gompertz-logistic gap of 0.039 is approximately two standard errors of the difference (SE $\approx$ 0.019), and the Gompertz default is the lowest-power, that is the mildly conservative, family of the four. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported in the direction anticipated: the breakpoint family tracks the logistic and the hyperbolic-tangent, not the Gompertz default.

An earlier draft of this manuscript reported essentially no family effect under either architecture. That result is now understood to have been an artifact of the interaction encoding rather than a property of trajectory family. In that draft the Architecture-A biomarker moderation was applied as a flat per-timepoint shift of magnitude $\beta_{bm}\sigma_{BR}$, independent of the response trajectory, so that the trajectory family, which enters only through the BR mean curve, could not affect the `bm:Dbc` interaction under the mean-moderation architecture any more than under the mean-independent the covariance-moderation architecture. Implementing the mean-moderation architecture in its faithful multiplicative form (section 4), in which the biomarker-dependent component scales with the response trajectory in keeping with the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`, the mean-moderation families separate at the few-percent level while the covariance-channel invariance under the covariance-moderation architecture is preserved, the latter being a genuine mean-independence property rather than an artifact.

6.2 Type I error across response-curve families

Type I error at $c_{bm} = 0$, percent $\pm$ MCSE:

Family	Mean	Covariance
Gompertz	1.9 ± 0.4	2.6 ± 0.5
Logistic	3.8 ± 0.6	2.2 ± 0.5
Hyperbolic tangent	2.7 ± 0.5	2.6 ± 0.5
Piecewise-linear breakpoint	2.8 ± 0.5	2.7 ± 0.5

All eight null cells produce empirical Type I error in $[0.019, 0.038]$, slightly conservative relative to the nominal 0.05 and consistent with the corCAR1 small-sample degree-of-freedom accounting at the prazosin-calibrated $N = 70$ and 8-timepoint short serial series. At the null the biomarker moderation term is zero, so the trajectory-scaling that distinguishes the Architecture-A families under the alternative is inactive here, and the null cells differ across families only by sampling noise. There is no family-specific inflation; the highest cell, logistic under the mean-moderation architecture at 0.038, is within two MCSE of the others. The conservatism is shared across all four families, supporting the interpretation that it reflects the corCAR1 d.f. mechanism rather than a trajectory-shape artifact, and it has appeared consistently across pmsimstats simulation runs at the reference parameters.

6.3 Bias of the biomarker-treatment interaction estimate

Across the eight alternative cells the mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ varies over a narrow range (roughly -0.230 to -0.239), a cell-to-cell spread of about 0.009 that is itself within the Monte Carlo noise of the estimator and shows no family-specific drift; the Gompertz and logistic cells that differ most in power have nearly identical mean estimates. The empirical SE of $\hat{\beta}_{bm:D_{bc}}$ is essentially constant across families (SD in the range 0.073 to 0.083 across cells) and roughly equal between architectures. That the Architecture-A power separation appears with negligible bias and near-constant empirical SE indicates it is a precision effect operating through the test statistic's sensitivity to the temporal profile of the signal, not a shift in the estimand or its sampling dispersion. The matched-anchor calibration holds the marginal effect size fixed across families by construction, so the residual family differences are differences in the temporal shape of the moderation, which is precisely the axis the design isolates.

7 Discussion

7.1 An architecture-dependent finding

The central finding is that the inferential consequence of trajectory-family choice is architecture-dependent. At the prazosin-calibrated reference cell (Hybrid OL+BDC, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), the four families span the qualitative range of plausible templates for monotone-saturating treatment-response trajectories: asymmetric (Gompertz), symmetric smooth (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint). Under the covariance-moderation architecture, where the interaction is carried by the BM-BR covariance and is therefore independent of the BR mean curve, the family is inert by construction. Under the mean-moderation architecture, where the interaction is a mean effect that, in its faithful multiplicative form, scales with the response trajectory, the families separate over a 0.039 power range and the Gompertz default is the lowest-power, that is the mildly conservative, family. We would not wish to overstate the Architecture-A effect; it is modest, on the order of two Monte Carlo standard errors at this sample size and effect size, and the conditions under which it would grow or shrink are discussed in section 6.3. But it is real and direction-consistent, and it overturns the apparent null reported in the earlier flat- moderation draft.

The pre-registered hypothesis that the piecewise-linear breakpoint family, with its non-differentiable feature, would behave qualitatively differently from the smooth families is not supported in the direction anticipated. The breakpoint family carries no distinctive power signature attributable to its non-smoothness; under the mean-moderation architecture it sits with the logistic and hyperbolic-tangent families, above the Gompertz default, and under the covariance-moderation architecture it is inert like the others. The operative axis is therefore not smoothness but the temporal profile of the biomarker-moderated signal: families whose response rises more gradually across the on-drug timepoints, and so carry more of the moderated signal at the later, more informative timepoints, yield marginally higher power than the front-loaded Gompertz curve, which reaches near-asymptote early.

7.2 Why the architectures differ

We see two mechanisms that, taken together, account for the architecture contrast, and we shall work through each in turn. The first is the exact mean-independence of the covariance channel under the covariance-moderation architecture; the second is the residual profile-sensitivity of the trajectory-scaled mean channel under the mean-moderation architecture, which the matched-anchor calibration and the time-averaging of the Wald test damp but do not eliminate.

The first mechanism damps the Architecture-A effect: calibration removes the amplitude differences, leaving only profile differences. The four families are matched at the asymptotic maximum and at a single anchor point ($t = 5$ weeks), and the trajectory-scaled moderation is normalized to a common mean on-drug magnitude. At matched marginal effect size, the residual within-trajectory variation in shape is small relative to the between-replicate variation in the `bm:Dbc` estimator. The calibration constraint is intentional in the simulation design: it isolates trajectory-shape effects from amplitude effects, so that the family-specific power differences we observe under the mean-moderation architecture can be attributed to the temporal profile of the moderation alone, and explains why that effect is on the order of a few percentage points rather than larger.

The second mechanism, also damping, is that the `bm:Dbc` Wald test averages over time. The continuous D_{bc} predictor multiplied by the biomarker integrates the trajectory's contribution across all on-drug timepoints, with weights determined by the predictor's exponential-decay structure. Trajectory-shape differences (where one family rises faster than another between timepoints) are partly averaged within the integration, so the Wald test is relatively, though not perfectly, insensitive to profile features. Under the mean-moderation architecture the averaging is incomplete: the residual profile difference survives into the test statistic and produces the observed few-percent power spread. Under the covariance-moderation architecture there is no profile term to survive, because the interaction does not enter the mean at all.

Mean-independence applies specifically to the covariance-moderation architecture: the covariance moderation is upstream of trajectory shape. Under MVN differential correlation, the biomarker-treatment information lives in the $\partial \mathrm{Cor}(BM, BR)/\partial D_{bc}$ structure, which is independent of the trajectory family that BR follows; the family enters only the BR mean, to which the interaction test is invariant. This is not a small-effect statement but an exact one, and we verify it directly: holding the random-number seed fixed across families, the four Architecture-B cells return bit-for-bit identical rejection decisions. The production-seed spread under the covariance-moderation architecture is therefore sampling noise, whereas the Architecture-A spread is a genuine, if modest, profile effect.

The combined effect is that trajectory-family choice is inferentially free under the covariance channel and carries a modest, mildly conservative, cost under the mean channel. The framework's modified Gompertz default is robust by exact construction under the covariance-moderation architecture, and under the mean-moderation architecture it is the lowest-power of the four families, which makes it a safe rather than an optimal default: a trial designer who adopts it under a mean-moderation DGP will, if anything, slightly under-state achievable power. We turn next to the conditions under which the Architecture-A effect would be expected to grow.

7.3 Conditions that would change the answer

Four conditions, none exercised in the present 16-cell factorial, would plausibly amplify the modest Architecture-A family effect, and each merits a brief comment. Because the covariance-moderation architecture is mean-independent, the first three conditions (sample size, effect size, carryover) bear on the mean-moderation architecture only; the fourth, non-monotone shape, is the only one that could in principle reach the covariance-moderation architecture, and then only by changing the covariance construction rather than the mean.

First, smaller sample size. At $N = 70$ the `bm:Dbc` test integrates over a moderately large within-subject sample; smaller N might amplify the local-shape sensitivity by concentrating the temporal information at fewer timepoints where the families differ more visibly.

Second, larger or smaller c_{bm}. The $c_{bm} = 0.45$ reference cell sits in a power regime where families have comparable signal-to-noise. Pushing c_{bm} to 0.30 (where power lies near 0.5) might surface family differences via different finite-sample sampling distributions, and it is possible that the tight clustering we observe is a feature of this particular region of the effect-size space.

Third, carryover-induced correlation decay. With $t_{1/2} > 0$, the `bm:Dbc`-mediated information channel is eroded differently for different trajectory shapes, because the integration weights interact with the off-drug exponential decay. The $t_{1/2} = 0$ slice exercised here suppresses this interaction entirely.

Fourth, non-monotone or biphasic shape envelopes. The four families we have examined share a monotone-saturating envelope. Families that peak and decay (biphasic) or oscillate would produce genuinely different trajectories with different `bm:Dbc` information content, and the Wald test would almost certainly respond to those differences.

Each condition is the natural axis of a follow-up sensitivity analysis. The 144-cell grid anticipated in the abstract ($N ×c_{bm} ×t_{1/2} ×$ family) is the canonical follow-up, and we recommend it as the production-grade extension that would close out the trajectory-family question at a level of rigor adequate for the methodological literature.

7.4 The prazosin/PTSD reference cell

For the motivating prazosin-PTSD application that drives the pmsimstats research program, the present result implies that sensitivity to the modified Gompertz assumption is small relative to sensitivity to the carryover specification (the subject of the companion manuscript at `analysis/report/02-carryover-sensitivity/`) and to the DGP architecture (the subject of `analysis/report/01-dgp-mean-moderation-vs-mvn/`). Of the three principal axes of sensitivity uncertainty in the pmsimstats simulation framework, trajectory family choice is, at the reference parameters, the least consequential by a substantial margin. It is not, however, entirely free: under the mean-moderation DGP the modified Gompertz default is the lowest-power family by a few percentage points, so a trial designer who adopts it is making a mildly conservative choice. Under the covariance-channel DGP the choice is inferentially free. We would suggest that sensitivity-analysis attention be directed chiefly to the carryover and DGP-architecture axes, where the inferential consequences are larger, while reporting at least one Gompertz-vs-alternative comparison whenever the operative DGP is mean moderation.

8 Conclusions

First, the modified Gompertz function is a defensible parametric template for monotone-saturating treatment-response trajectories in N-of-1 simulation, with an architecture-dependent qualification. At the prazosin/PTSD reference cell with matched-anchor calibration, the four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) produces exactly equivalent inference under the covariance-channel architecture, where the interaction is mean-independent, and a modest 0.039 power spread under the mean-moderation architecture, where the biomarker moderation rides the response trajectory. In the latter case the Gompertz default is the lowest-power of the four families, which makes it a mildly conservative rather than an optimal default. The pre-registered hypothesis that the piecewise-linear breakpoint family would behave qualitatively differently from the smooth families is not supported in the direction anticipated.

Second, the finding is methodologically informative precisely because it is architecture-dependent. It implies that the `bm:Dbc` Wald test is exactly robust to trajectory-family choice when the interaction is encoded in the covariance, and only mildly sensitive to it, at the level of a few percentage points of power, when the interaction is a trajectory-scaled mean effect. We cannot yet establish at what sample sizes, effect sizes, or carryover levels the Architecture-A effect would grow; that question is deferred to the 144-cell follow-up. Trial designers using the pmsimstats framework can adopt the modified Gompertz default with confidence under a covariance-channel DGP, and with the understanding that it is mildly conservative under a mean-moderation DGP, while directing sensitivity-analysis attention to the carryover and DGP-architecture axes that have larger inferential consequences.

Third, the 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is the natural follow-up, and we regard it as the work that would be needed to close the trajectory-family question at a level of rigor adequate for the methodological literature. The present focused 16-cell factorial addresses the central scientific question with statistical clarity; the extended grid would map the boundary conditions at which trajectory-family choice begins to matter.

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

The simulation driver `analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R` sets `set.seed(20260617)` and assigns each of the 16 cells a distinct seed offset, so replicate streams are independent across cells but reproducible. The trajectory family is routed through the data-generating process via the `br_family` argument to `buildSigma`, and the Architecture-A moderation is trajectory-scaled via the `moderation_scaling = "trajectory"` argument to `generateData` (both default to the original behavior, so all other pmsimstats runs are unchanged). Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; Morris ADEMP cell-level summaries are at the companion `07-gompertz-faithful-summary.txt`. The pre-registered ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`, and deviations from it are recorded in `02-deviations-log.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/07-gompertz-evaluation/`

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